EXPERIMENT - 19

19. Implement the Erosion Morphological operations technique using Open CV in python.

Aim

To implement the Erosion Morphological Operation using OpenCV in Python to reduce the boundaries of foreground objects in an image.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply erosion using cv2.erode().
7. Display the original image and eroded output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = np.ones((5, 5), np.uint8)

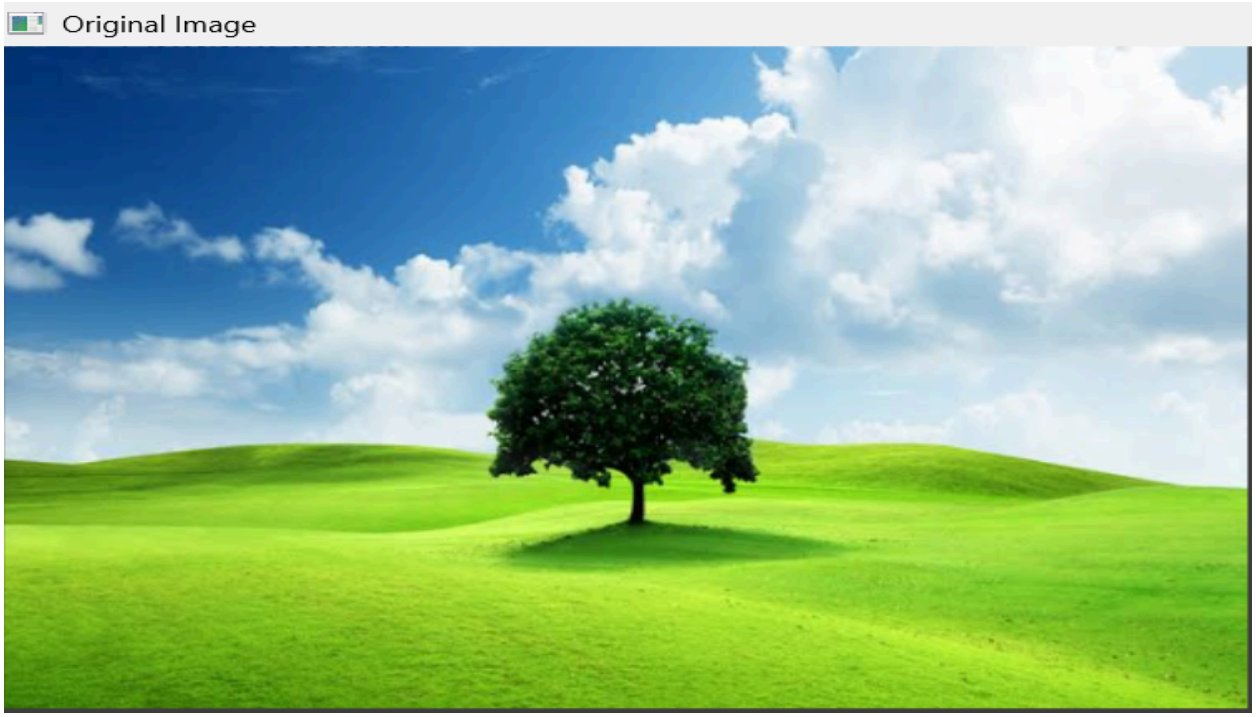
    erosion = cv2.erode(gray, kernel, iterations=1)

    cv2.imshow("Original Image", img)
    cv2.imshow("Erosion Output", erosion)

    cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input Image



Output Image



Result

The Erosion Morphological Operation was successfully implemented using OpenCV in Python.